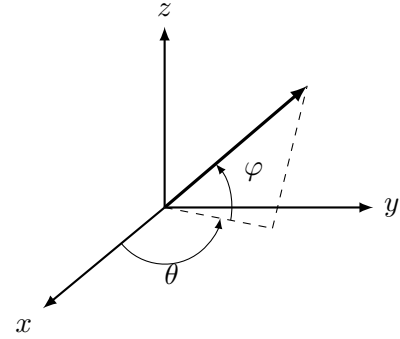


# Sensor location

$$\begin{cases} \varphi = [-90^\circ, 90^\circ] & \text{仰角} \\ \theta = [0^\circ, 360^\circ] & \text{方位角} \\ \beta = 11.01^\circ \text{ (上磁極)}, 11.49^\circ \text{ (下磁極)} \\ \alpha = \text{旋轉角} \end{cases}$$



## 1. 從 WP 中心到極尖

$$\hat{e}_1(\varphi, \theta) = \begin{bmatrix} \cos \varphi \cos \theta \\ \cos \varphi \sin \theta \\ \sin \varphi \end{bmatrix} \quad \varphi_1 = \begin{cases} +35.2644^\circ & \text{上磁極} \\ -35.2644^\circ & \text{下磁極} \end{cases}$$

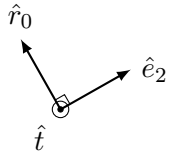
$\hat{e}_1$  用  $\varphi_1$ 、 $\hat{e}_2$  用  $\varphi_2$ ，兩者不是同一個值。

## 2. 從極尖到指定位置

$$\hat{e}_2(\varphi, \theta) = \begin{bmatrix} \cos \varphi \cos \theta \\ \cos \varphi \sin \theta \\ \sin \varphi \end{bmatrix} \quad \varphi_2 = \begin{cases} 36.5895^\circ & \text{上磁極} \\ 0^\circ & \text{下磁極} \end{cases}$$

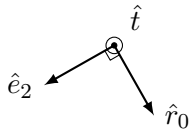
## 3.

### (1) 上磁極



$$\hat{r}_0(\varphi, \theta) = \hat{e}_2(\varphi + 90^\circ, \theta) = \begin{bmatrix} -\sin \varphi \cos \theta \\ -\sin \varphi \sin \theta \\ \cos \varphi \end{bmatrix}, \quad \hat{t} = \hat{e}_2 \times \hat{r}_0$$

### (2) 下磁極



$$\hat{r}_0(\varphi, \theta) = \hat{e}_2(\varphi - 90^\circ, \theta) = \begin{bmatrix} \sin \varphi \cos \theta \\ \sin \varphi \sin \theta \\ -\cos \varphi \end{bmatrix}, \quad \hat{t} = \hat{e}_2 \times \hat{r}_0$$

$$(3) \quad \hat{r}(\alpha, \varphi, \theta) = \cos \alpha \cdot \hat{r}_0 + \sin \alpha \cdot \hat{t} \quad \begin{cases} \text{上磁極} & \alpha = [0^\circ, 360^\circ] \\ \text{下磁極} & \alpha = [-90^\circ, 90^\circ] \end{cases}$$

4.

$$\begin{aligned}\hat{n} &= \hat{e}_2(\varphi + 90^\circ + \beta, \theta) \quad (\text{上極}), \quad \hat{n} = \hat{e}_2(\varphi - 90^\circ - \beta, \theta) \quad (\text{下極}) \\ \implies \quad \hat{n}(\alpha, \varphi, \theta) &= \cos \beta \cdot \hat{r}(\alpha, \varphi, \theta) - \sin \beta \cdot \hat{e}_2(\varphi, \theta)\end{aligned}$$

## 5. 底面位置

(1) 上極

$$\begin{aligned}\vec{P}_{(\text{mm})} &= 0.5 \cdot \hat{e}_1(\varphi, \theta) + [L \cos \beta] \hat{e}_2(\varphi, \theta) \\ &\quad + [0.03922 + (L - 0.03198) \sin \beta] \cdot \hat{r}(\alpha, \varphi, \theta) + 0.41 \cdot \hat{n}\end{aligned}$$

(2) 下極

$$\begin{aligned}\vec{P}_{(\text{mm})} &= 0.5 \cdot \hat{e}_1(\varphi, \theta) + [L \cos \beta] \hat{e}_2(\varphi, \theta) \\ &\quad + [0.03922 + (L - 0.03280) \sin \beta] \cdot \hat{r}(\alpha, \varphi, \theta) + 0.41 \cdot \hat{n}\end{aligned}$$